

MARCO FALTELLI

CONTACT INFORMATION

ENEA - Frascati Research Center

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Google Scholar: scholar.google.it/citations?user=joXyxfoAAAAJ&hl

RESEARCH INTERESTS

- **Topics:** High-Performance Computing, Parallel Computing, Data Centers, Operative Systems, Distributed Systems, Network Drivers, Machine Learning, Fast Packet Processing
- **Hardware:** FPGA, GPU, SmartNIC, HW-SW co-design, speculative HW

EDUCATION

University of Rome “Tor Vergata”

PhD in Computer Science, Control and GeoInformation

Rome, Italy

November 2019 - June 2023

Thesis: *Towards resilient and effective network services*

Advisors: Prof. Giuseppe Bianchi, Prof. Francesco Quaglia

Final Evaluation: *Laude* (top 5% quality)

Master’s Degree in Computer Engineering

October 2017 - October 2019

Thesis: *A state-machine based platform for portable transport protocols*

Advisors: Prof. Giuseppe Bianchi, Prof. Francesco Quaglia

Final Evaluation: 110/110 *cum laude*

Bachelor’s Degree in Computer Engineering

October 2014 - October 2017

Final Evaluation: 110/110

TECHNICAL STRENGTHS

Programming Languages

Programming Skills

HPC

Networking

Software & Tools

Frameworks & Services

Operative Systems

C, C++, Java, Python, Javascript, Linux Kernel, Android, SQL, HTML
non-blocking algorithms, code optimization, compilers, profilers, debuggers
MPI, OpenMP, RDMA, FPGA (Vitis), GPU (CUDA), SLURM
SDN, NFV, DPDK, eBPF/XDP, Network Drivers, P4
IDE, git, Matlab, R, MS Office, LaTeX
AWS, ML (Tensorflow, Keras), Big Data (Hadoop, Flink)
Linux, Windows 10, OS X

WORK EXPERIENCE

ENEA

Researcher

Frascati, Italy

December 2023 - present

- Researcher in the TERIN-ICT-HPC department with a focus on HPC and Data Centers
- High-performance task acceleration using FPGA and GPU (**TEXTAROSSA** EU project)
- Ongoing work in bringing HPC to the cloud (**IPCEI-CIS** EU project - ENEA WS1 leader)

University of Rome “Tor Vergata”

Technologist

Rome, Italy

March 2023 - December 2023

- Parallel computing and code optimization methods for the National Centre for HPC, Big Data and Quantum Computing - NextgenerationEU project

Microsoft

Research Intern

Cambridge, United Kingdom

July 2022 - Oct. 2022

- High-performing optimization methods for robots path planning in the Silica project

NEC Laboratories Europe

Research Intern

Heidelberg, Germany

October 2021 - Dec. 2021

- Acceleration methods for 5G edge computing scenarios

CNIT

Student Researcher

Rome, Italy

October 2017 - December 2019

- Developer and maintainer for FlowBlaze and XTRA, open source projects for fast and stateful packet processing in hardware. Projects conducted under the H2020-5G PICTURE grant.
- Worked on a railway use-case scenario for ensuring session continuity as part of the H2020-5G PICTURE project.

University of Rome Tor Vergata

Computer Science tutor

Rome, Italy

Nov. 2020 - March 2021

AWARDS & HONORS

- Winner of a 2022 **Microsoft Research PhD Fellowship**
- Participation to the 2023 **CERN School of Computing**
- “*A Fully Portable TCP Implementation Using XFSMs*” has earned the 3rd place in the ACM SIGCOMM 2018 Student Research Competition (Undergraduate class).
- Winner of a scholarship from the Italian Ministry of Education for outstanding student results in 2017.

SERVICE

Reviewer for *International Conference on Supercomputing 2026* (ACM), *Scientific Reports* (Nature) and *International Journal of Supercomputing* (Springer)

PUBLICATIONS

- L. Piarulli, M. Faltelli, D. Pleiter, K. Sivalingam, D. Zhang, K. Zhao, M. Turisini, F. Iannone, A. Artigiani, and D. De Sensi. Characterizing the Impact of Congestion in Modern HPC Interconnects. *ISC High Performance 2026*

- M. Faltelli, A. Peloso, F. Iannone, M. Fois, M. Celino, and G. Ponti. CLIC (Cloud In Cresco): towards HPC/HPDA-as-a-Service. In *GARR Conference 2025*, 2025
- P. Palazzari, M. Faltelli, F. Iannone and F. Pascarella. Fast and Energy-Efficient N-Body Simulation on FPGA through High-Level Synthesis. *2025 33rd Euromicro International Conference on Parallel, Distributed, and Network-Based Processing (PDP)*, 2025
- P. Palazzari, M. Faltelli, and F. Iannone. FIPLib: An Image Processing Library for FPGAs Using High-Level Synthesis. *International Journal of Parallel Programming*, 53(2):13, 2025
- M. Faltelli, G. Belocchi, F. Quaglia, and G. Bianchi. COREC: Concurrent non-blocking single-queue receive driver for low latency networking. *Computer Networks*, 258:110982, 2025
- M. Faltelli, G. Belocchi, F. Quaglia, S. Pontarelli, and G. Bianchi. Metronome: adaptive and precise intermittent packet retrieval in DPDK (extended version). In *IEEE/ACM Transactions on Networking*, Oct. 2022
- M. Faltelli, G. Belocchi, F. Quaglia, S. Pontarelli, and G. Bianchi. Metronome: adaptive and precise intermittent packet retrieval in DPDK. In *ACM CoNEXT 2020*
- G. Bianchi, M. Welzl, A. Tulumello, F. Gringoli, G. Belocchi, M. Faltelli, and S. Pontarelli. XTRA: Towards Portable Transport Layer Functions. *IEEE Transactions on Network and Service Management*, Dec 2019
- V. Bruschi, M. Faltelli, A. Tulumello, S. Pontarelli, F. Quaglia, and G. Bianchi. Offloading online MapReduce tasks with stateful programmable data planes. In *IEEE NETPROC 2020*
- G. Bianchi, M. Faltelli, and V. Bruschi. Back to the Future: Towards Hardware "Netputing" Architectures. In *IEEE MedComNet 2020*
- G. Bianchi, M. Welzl, A. Tulumello, G. Belocchi, M. Faltelli, and S. Pontarelli. A Fully Portable TCP Implementation Using XFSMs. In *ACM SIGCOMM 2018 Posters and Demos*
- M. Spaziani Brunella and M. Faltelli. Exploiting Foreshadow-VMM. Technical report, CNIT, 2019